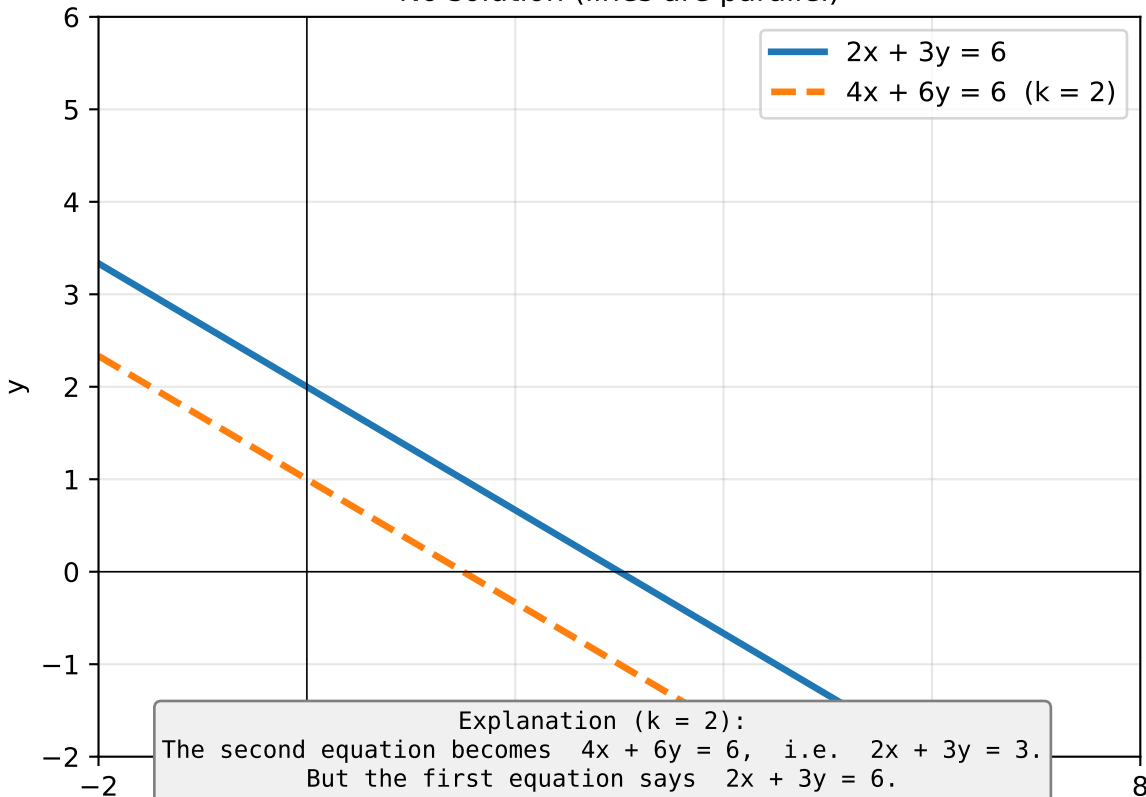


$k = 2 \rightarrow \text{rank}(A) = 1, \text{rank}([A|B]) = 2$

No solution (lines are parallel)



Explanation ($k = 2$):

The second equation becomes $4x + 6y = 6$, i.e. $2x + 3y = 3$.

But the first equation says $2x + 3y = 6$.

Since $3 \neq 6$, the lines are parallel and never meet.

$\text{rank}(A) = 1 \neq \text{rank}([A|B]) = 2 \rightarrow$ no solution.